

SECTION 23 25 00 - HVAC WATER TREATMENT

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. By-pass (pot) feeder.
- B. Drip feeder.
- C. Solution metering pump.
- D. Solution tanks.
- E. Agitator.
- F. Liquid level switch.
- G. Conductivity controller.
- H. Water meter.
- I. Solenoid valves.
- J. Test equipment.

1.02 RELATED REQUIREMENTS

- A. Section 23 09 13 - Instrumentation and Control Devices for HVAC.
- B. Section 23 21 13 - Hydronic Piping.

1.03 REFERENCE STANDARDS

- A. UL (DIR) - Online Certifications Directory Current Edition.

1.04 SUBMITTALS

- A. Product Data: Provide chemical treatment materials, chemicals, and equipment including electrical characteristics and connection requirements.
- B. Shop Drawings: Indicate system schematic, equipment locations, and controls schematics, electrical characteristics and connection requirements.

PART 2 PRODUCTS

2.01 REGULATORY REQUIREMENTS

- A. Comply with applicable codes for addition of non-potable chemicals to building mechanical systems and to public sewage systems.
- B. Comply with UL (DIR) requirements.
- C. Provide certificate of compliance from Authority Having Jurisdiction indicating approval of installation.

2.02 BY-PASS (POT) FEEDER

- A. 2 quart (1.9 L) quick opening cap for working pressure of 175 psi (1200 kPa).

2.03 DRIP FEEDER

- A. Plastic reservoir with coil of capillary tubing with probe, weight, charging syringe, and clip.

2.04 SOLUTION METERING PUMP

- A. Positive displacement, diaphragm pump with adjustable flow rate, thermoplastic construction, continuous-duty fully enclosed electric motor and drive, and built-in relief valve.

2.05 SOLUTION TANKS

- A. 30 gallon (114 L) capacity, polyethylene, self-supporting, 1 gallon (3.8 L) graduated markings; molded fiberglass cover with recess for mounting pump, agitator, and liquid level switch.

2.06 AGITATOR

- A. Totally enclosed electric motor, cast iron clamp and motor mount, 1/2 inch (13 mm) diameter coated Type 316 stainless steel propeller.

2.07 LIQUID LEVEL SWITCH

- A. Polypropylene housing with integrally mounted PVC air trap, receptacles for connection to metering pump, and low level alarm.

2.08 CONDUCTIVITY CONTROLLER

- A. Packaged monitor controller with solid state circuiting, five percent accuracy, linear dial adjustment, built-in calibration switch, on-off switch and light, control function light, output to control circuit and recorder.

2.09 WATER METER

- A. Displacement type cold water meter with sealed, tamper-proof magnetic drive, impulse contact register, single pole, double throw dry contact switch.

2.10 SOLENOID VALVES

- A. Forged brass body globe pattern, normally open or closed as required, explosion-proof and watertight solenoid enclosure, and continuous duty coil.

2.11 TEST EQUIPMENT

- A. Provide white enamel test cabinet with local and fluorescent light, capable of accommodating 4 - 10 ml zeroing titrating burettes and associated reagents.

END OF SECTION 23 25